

Correcting an Infinite-Horizon Coxian Threshold Argument and a Conditional Finite-Horizon Result for the Refined Scheme

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Abstract

This note separates two different assertions about the refined learning scheme of Goldsztajn, Borst and van Leeuwaarden. First, it gives a corrected proof that the literal infinite-horizon statement sometimes extracted from their paper is false: in any fixed finite system with positive arrival rate, the refined threshold is almost surely not eventually constant. This is not a contradiction of the authors' finite-horizon many-server theorems. Second, the note gives a genuine finite-horizon partial result for the Coxian/refined case. Let $\ell = \lfloor \rho \rfloor_\Delta$ and let

$$\beta := \max_{1 \leq i \leq r} (1 - p_i) \mu_i$$

be the largest instantaneous system-exit rate of a single Coxian phase. Under the additional slow-exit condition $\beta \ell < \lambda$, the missing threshold-equilibrium part of the Coxian/refined extension holds on every fixed finite time horizon: after a deterministic time, $\ell_n(t) = \ell$ for all large n , uniformly on the horizon. Thus the paper below proves a conditional solution for this subclass of Coxian laws, but not the full Coxian/refined open extension.

1 Status of the result

The uploaded draft contained a correct negative idea, but it was aimed at a statement stronger than the one used in the paper. The corrected conclusions are as follows.

- (a) The literal assertion that there is one finite N and one finite time σ such that $\ell_n(t) = \lfloor \rho \rfloor_\Delta$ for every $n \geq N$ and every $t \geq \sigma$ is false when $\lambda > 0$. This is proved in [Section 3](#).
- (b) The actual finite-horizon Coxian/refined problem is not fully solved here. The new positive result is conditional: if the largest Coxian phase-exit rate β satisfies $\beta \lfloor \rho \rfloor_\Delta < \lambda$, then the refined threshold reaches the correct equilibrium on every fixed finite horizon. This is proved in [Section 4](#).

The second statement is genuine partial progress toward the Coxian/refined extension mentioned in Section 4.3 of [\[1\]](#). It is not the full extension, because a general Coxian distribution may have a phase with exit rate larger than $\lambda / \lfloor \rho \rfloor_\Delta$.

2 Model

We use the notation of [\[1\]](#). There are n identical server pools, each with infinitely many servers. Tasks arrive according to a Poisson process of rate $n\lambda$, where $\lambda > 0$ is constant. Service requirements are Coxian with r phases. A task in phase i completes that phase after an exponential time of

rate μ_i ; after phase i it leaves the system with probability $1 - p_i$ and moves to phase $i + 1$ with probability p_i , where $p_r = 0$. The mean service time is denoted by $1/\mu$, and the offered load per pool is

$$\rho := \lambda/\mu.$$

For a pool occupancy process $X_k^n(t)$, define the occupancy tail fractions

$$q_n(t, j) := \frac{1}{n} \sum_{k=1}^n \mathbf{1}\{X_k^n(t) \geq j\}, \quad j \geq 0.$$

The refined threshold $\ell_n(t)$ takes values in $\Delta\mathbb{Z}_+$, where $\Delta \geq 1$, and $h_n(t) := \ell_n(t) + \Delta$. At each arrival epoch, the incoming task is dispatched by the threshold rule:

- if some pool has fewer than ℓ_n tasks, send the task to such a pool;
- otherwise, if some pool has fewer than h_n tasks, send the task to a pool with occupancy in $[\ell_n, h_n)$;
- otherwise, send it to a pool with at least h_n tasks.

After dispatching that task, the refined learning rule updates the threshold using the pre-arrival state:

- if $q_n(T-, \ell_n(T-)) \leq \alpha_n$, decrease ℓ_n by Δ , with reflection at zero;
- if $q_n(T-, \ell_n(T-)) = 1$ and $nq_n(T-, h_n(T-)) \geq n - 1$, increase ℓ_n by Δ .

The scaling assumption on the refinement parameter is the same one used for the refined scheme in [1].

Assumption 2.1 (Refinement scaling). *There is $\gamma_0 \in (0, 1/2)$ such that*

$$\alpha_n \rightarrow 1, \quad \liminf_{n \rightarrow \infty} n^{\gamma_0} (1 - \alpha_n) > 0.$$

The normalized aggregate number of tasks is

$$u_n(t) := \sum_{j \geq 1} q_n(t, j).$$

We use the following standard many-server initial condition. It is slightly more explicit than needed, but it keeps the finite-horizon theorem self-contained.

Assumption 2.2 (Initial state and aggregate law of large numbers). *There is a deterministic $B_0 < \infty$ such that all pools initially have at most B_0 tasks and $\ell_n(0) \leq B_0$ for all large n . The aggregate Coxian infinite-server process satisfies, on the graphical construction under consideration, the compact-time law of large numbers*

$$\sup_{0 \leq t \leq T} |u_n(t) - u(t)| \longrightarrow 0 \quad \text{a.s. for every } T < \infty,$$

where u is the deterministic $M/PH/\infty$ fluid limit with arrival rate λ and mean service time $1/\mu$. In particular, $u(t) \rightarrow \rho = \lambda/\mu$ as $t \rightarrow \infty$.

3 The literal infinite-horizon formulation is false

The first result repairs the uploaded draft's negative theorem. It is deliberately stated for fixed n . Therefore it refutes only the literal infinite-horizon quantifier order, not the finite-horizon many-server statement.

Lemma 3.1 (Empty epochs). *For each fixed n and $\lambda > 0$, starting from any finite state, the whole system is empty at arbitrarily large times almost surely.*

Proof. Ignore the assignment of jobs to pools and record only the total number of jobs in each Coxian phase. This is the usual $M/PH/\infty$ Markov chain with arrival rate $n\lambda$. If S is a service time, the stationary probability that all phase counts are zero is $\exp(-n\lambda\mathbb{E}S) > 0$. The phase-count chain is irreducible on the communicating class reached from a finite initial state and is positive recurrent. Hence every state of positive stationary probability, in particular the all-empty state, is recurrent. Therefore the all-empty state is visited infinitely many times almost surely. At such times every pool is empty. $\square$

Lemma 3.2 (A burst from threshold zero). *Fix $n \geq 1$. Suppose that at a stopping time τ the system is empty and $\ell_n(\tau) = 0$. Conditional on the past up to τ , there is a constant $p_{n,\Delta} > 0$, independent of the past, such that with conditional probability at least $p_{n,\Delta}$ the threshold increases to Δ before the system next becomes empty.*

Proof. Let $M = n\Delta$ and choose $a > 0$. Let B be the event that the first M arrivals after τ occur before time $\tau + a$ and that the service times of those M tasks all exceed a . By the strong Markov property and independence of future arrivals and services,

$$\mathbb{P}(B) = \mathbb{P}\{\Gamma(M, n\lambda) < a\} \mathbb{P}\{S > a\}^M > 0,$$

where $\Gamma(M, n\lambda)$ is the time of the M th arrival in a Poisson process of rate $n\lambda$.

On B no service completion occurs before the M th arrival. Starting empty with threshold 0, the dispatching rule sends every arrival to a pool with fewer than Δ tasks until all pools have Δ tasks. Just before the M th arrival, there are $n\Delta - 1$ tasks, no pool has more than Δ tasks, and at least $n - 1$ pools have at least Δ tasks. The refined increase condition is therefore satisfied at the M th arrival, and after that arrival the threshold is Δ . This occurs before the next empty epoch. $\square$

Theorem 3.3 (No fixed finite system has an eventual constant threshold). *For every fixed $n \geq 1$ and every $L \in \Delta\mathbb{Z}_+$,*

$$\mathbb{P}(\exists \sigma < \infty \text{ such that } \ell_n(t) = L \text{ for all } t \geq \sigma) = 0.$$

Proof. For $m \in \mathbb{N}$ let

$$A_L(m) := \{\ell_n(t) = L \text{ for all } t \geq m\}.$$

It suffices to prove $\mathbb{P}(A_L(m)) = 0$ for each fixed m .

First suppose $L > 0$. On $A_L(m)$, by Lemma 3.1 there is an all-empty time $E \geq m$. Let $T > E$ be the next arrival epoch. Immediately before T , the system is empty, so $q_n(T-, L) = 0 \leq \alpha_n$. The refined rule decreases the threshold after the arrival, contradicting $A_L(m)$. Hence $\mathbb{P}(A_L(m)) = 0$ for all $L > 0$.

Now suppose $L = 0$. On $A_0(m)$ the threshold equals zero at every empty epoch after m . Let $E_1, E_2, \dots$ be successive empty epochs after m , which exist almost surely by Lemma 3.1. At each E_k , conditional on the past and on $A_0(m)$, the burst event from Lemma 3.2 has probability at least

$p_{n,\Delta} > 0$. Therefore the conditional probability that no burst occurs in the first K empty cycles is at most $(1 - p_{n,\Delta})^K$, which tends to zero. Almost surely some burst occurs, and then the threshold increases to Δ , contradicting $A_0(m)$. Thus $\mathbb{P}(A_0(m)) = 0$. $\square$

Corollary 3.4 (The literal many-server half-line statement is false). *For any deterministic $\ell^* \in \Delta\mathbb{Z}_+$,*

$$\mathbb{P}\left(\exists N < \infty, \exists \sigma < \infty : \ell_n(t) = \ell^* \text{ for all } n \geq N \text{ and all } t \geq \sigma\right) = 0.$$

Proof. For a fixed value of N , the event inside the probability implies eventual constancy of the single fixed- N process ℓ_N , which has probability zero by [Theorem 3.3](#). Take the countable union over N . $\square$

Remark 3.5. *This is compatible with a finite-horizon theorem. A statement of the form*

$$\lim_{n \rightarrow \infty} \sup_{t \in [\sigma, T]} |\ell_n(t) - \lfloor \rho \rfloor_\Delta| = 0 \quad \text{for each fixed } T < \infty$$

allows the required lower bound on n to depend on T . [Theorem 3.4](#) rules out only a single finite N that works on the whole half-line $[\sigma, \infty)$.

4 A finite-horizon partial result for the Coxian/refined problem

We now turn to the intended finite-horizon problem. The argument below proves the missing threshold conclusion under an additional condition on the Coxian exit rates.

Define

$$\ell := \lfloor \rho \rfloor_\Delta, \quad \beta := \max_{1 \leq i \leq r} (1 - p_i) \mu_i.$$

Thus β is the largest rate at which one task, in a specified Coxian phase, can leave the system. Phase changes that do not leave the system do not change a pool's occupancy and therefore are irrelevant to the column estimates below.

Theorem 4.1 (Conditional Coxian/refined threshold equilibrium). *Assume [Theorems 2.1](#) and [2.2](#) and $\rho \notin \Delta\mathbb{Z}_+$. If*

$$\beta\ell < \lambda, \tag{1}$$

then there is a deterministic $\sigma < \infty$, independent of the final horizon T , such that for every fixed $T > \sigma$ there is an event of probability one on which

$$\ell_n(t) = \ell \quad \text{for all } t \in [\sigma, T]$$

for all sufficiently large n . Consequently,

$$\lim_{n \rightarrow \infty} \sup_{t \in [\sigma, T]} |\ell_n(t) - \ell| = 0 \quad \text{almost surely.}$$

4.1 Pathwise estimates

The proof uses two elementary consequences of the graphical construction. The first is the familiar no-saturation property of the refined rule.

Lemma 4.2 (No saturation of the current upper level). *For every $\varepsilon > 0$ and every fixed $T < \infty$, almost surely, for all sufficiently large n ,*

$$q_n(t, h_n(t)) < 1 \quad \text{for every } t \in [\varepsilon, T].$$

Proof. If $q_n(t, h_n(t)) < 1$ at some time, the refined update rule preserves this strict inequality. Indeed, if an arrival could make all pools have at least the current upper level, then just before that arrival at least $n - 1$ pools had at least the upper level and all pools had at least the current threshold, so the refined rule would immediately increase the threshold; with the new upper level the strict inequality again holds. Departures can only help.

It remains to handle the possibility that the initial upper level is saturated. Since all initial occupancies and initial thresholds are bounded by B_0 , after at most $B_0 + 2$ consecutive arrivals the threshold has been increased enough that the current upper level exceeds $B_0 + 1$, while fewer than $B_0 + 2$ pools can have been raised above the initial maximum by those arrivals. For all $n > B_0 + 2$, not every pool can then have reached the new upper level. Since a Poisson process of rate $n\lambda$ has more than $B_0 + 2$ arrivals by time ε for all large n almost surely, the strict inequality holds throughout $[\varepsilon, T]$ for all large n . $\square$

For $J \geq 1$, let $C_{n,J}(t)$ be the normalized number of tasks in the first J columns of the occupancy diagram,

$$C_{n,J}(t) := \sum_{i=1}^J q_n(t, i) = \frac{1}{n} \sum_{k=1}^n \min\{X_k^n(t), J\}.$$

Let $D_{n,J}(s, t)$ be the number of service exits during $(s, t]$ from pools whose pre-exit occupancy is at most J . Such exits are the only service exits that can decrease $C_{n,J}$.

Lemma 4.3 (Arrival/departure strong estimate). *Fix $J < \infty$ and $T < \infty$. On an event of probability one there is a deterministic sequence $e_n = e_n(J, T)$ with*

$$e_n = o(1 - \alpha_n)$$

such that, for all sufficiently large n and all $0 \leq s \leq t \leq T$,

$$\frac{A_n(t) - A_n(s)}{n} \geq \lambda(t - s) - e_n, \tag{2}$$

$$\frac{D_{n,J}(s, t)}{n} \leq \beta J(t - s) + e_n. \tag{3}$$

Here A_n is the arrival counting process.

Proof. For arrivals this follows from the strong law, or the law of the iterated logarithm, for the Poisson process $A_n(t) = Y_n(n\lambda t)$: for every $\gamma < 1/2$,

$$\sup_{0 \leq t \leq T} \left| \frac{A_n(t)}{n} - \lambda t \right| = o(n^{-\gamma}) \quad \text{a.s.}$$

Choose $\gamma = \gamma_0$ from [Theorem 2.1](#); the assumption on α_n makes this error $o(1 - \alpha_n)$.

For departures, the total instantaneous rate of exits from pools with at most J tasks is at most $n\beta J$: each such pool contains at most J tasks and each task exits the system at rate at most β . By the random time-change representation,

$$D_{n,J}(0, t) = Y'_n \left(\int_0^t R_{n,J}(v) dv \right), \quad 0 \leq R_{n,J}(v) \leq n\beta J,$$

for a unit-rate Poisson process Y'_n . Hence, uniformly over $0 \leq s \leq t \leq T$,

$$\frac{D_{n,J}(s, t)}{n} \leq \beta J(t - s) + \frac{2}{n} \sup_{0 \leq u \leq n\beta JT} |Y'_n(u) - u|.$$

The last term is $o(n^{-\gamma_0})$ almost surely, and therefore $o(1 - \alpha_n)$. $\square$

Lemma 4.4 (Positive drift below a filled column). *Fix $J \in \Delta\mathbb{Z}_+$ with $J \geq 1$. Work on the event where Lemmas 4.2 and 4.3 hold. Suppose that, on an interval $[s, t] \subset [0, T]$,*

$$\ell_n(v) \leq J, \quad q_n(v, h_n(v)) < 1, \quad q_n(v, J) < 1 \quad \text{for all } v \in [s, t].$$

Then, for all large n ,

$$C_{n,J}(t) - C_{n,J}(s) \geq (\lambda - \beta J)(t - s) - 2e_n(J, T). \quad (4)$$

Proof. During the interval in question, every arrival is routed to a pool with fewer than J tasks. Indeed, if a pool has fewer than $\ell_n \leq J$ tasks, this is immediate. Otherwise, all pools have at least ℓ_n tasks. Since the current upper level is not saturated, some pool has fewer than $h_n = \ell_n + \Delta$ tasks; the dispatching rule then sends the arrival to a pool with occupancy below h_n . If $\ell_n < J$, then $h_n \leq J$ because both ℓ_n and J are multiples of Δ . If $\ell_n = J$, then $q_n(v, J) < 1$ gives a pool below J and the first dispatching case applies. Thus every arrival increases $C_{n,J}$ by $1/n$.

Only service exits from pools with pre-exit occupancy at most J can decrease $C_{n,J}$, and each such exit decreases it by at most $1/n$. Therefore

$$C_{n,J}(t) - C_{n,J}(s) \geq \frac{A_n(t) - A_n(s)}{n} - \frac{D_{n,J}(s, t)}{n}.$$

Applying Lemma 4.3 gives (4). □

4.2 Proof of the conditional finite-horizon theorem

Proof of Theorem 4.1. Let $\ell = \lfloor \rho \rfloor_\Delta$. Since $\rho \notin \Delta\mathbb{Z}_+$,

$$\ell < \rho < \ell + \Delta.$$

Choose $\varepsilon_0 > 0$ such that $\rho + 2\varepsilon_0 < \ell + \Delta$. Since the Coxian infinite-server limit satisfies $u(t) \rightarrow \rho$, choose a deterministic time t_0 such that

$$u(t) \leq \rho + \varepsilon_0 < \ell + \Delta - \varepsilon_0 \quad \text{for every } t \geq t_0.$$

Fix a finite horizon $T > t_0 + 1$. By uniform convergence of u_n to u on $[0, T]$, almost surely, for all large n ,

$$u_n(t) \leq \ell + \Delta - \varepsilon_0 \quad \text{for every } t \in [t_0, T]. \quad (5)$$

Consequently, for every $j \geq \ell + \Delta$ and every $t \in [t_0, T]$,

$$q_n(t, j) \leq \frac{u_n(t)}{j} \leq \frac{\ell + \Delta - \varepsilon_0}{\ell + \Delta} =: 1 - \delta_0, \quad (6)$$

where $\delta_0 > 0$. For all large n , $1 - \delta_0 < \alpha_n$ and $1 - \delta_0 < 1 - 1/n$.

We first prove the eventual upper bound $\ell_n \leq \ell$. On compact time intervals the threshold is bounded for all large n : after the no-saturation time this follows from the fact that an increase to a level j requires $q_n(j) \geq 1 - 1/n$, and hence $j(1 - 1/n) \leq u_n$; before that time the bounded initial condition and the argument in Lemma 4.2 give a deterministic finite bound. Let $B < \infty$ be such a bound on $[0, t_0 + 1]$. If $\ell_n(t) \geq \ell + \Delta$ at a time $t \in [t_0, T]$, then $q_n(t, \ell_n(t)) < \alpha_n$ by (6), so the threshold decreases at the next arrival. Also, while $\ell_n(t) \geq \ell$, threshold increases are impossible, because $h_n(t) \geq \ell + \Delta$ and (6) gives $nq_n(t, h_n(t)) < n - 1$. Since the arrival rate is $n\lambda$, the interval

$[t_0, t_0 + 1]$ contains more than $B/\Delta + 1$ arrivals for all large n almost surely. Therefore, for all large n ,

$$\ell_n(t) \leq \ell \quad \text{for every } t \in [t_0 + 1, T]. \quad (7)$$

If $\ell = 0$, this proves the theorem, with any $\sigma > t_0 + 1$.

Assume now that $\ell \geq 1$. Put

$$a_* := \lambda - \beta\ell > 0, \quad R := \frac{\ell + 1}{a_*}, \quad t_1 := t_0 + 1.$$

By Lemma 4.2, increasing t_1 by an arbitrarily small deterministic amount if necessary, we may also assume that $q_n(t, h_n(t)) < 1$ on $[t_1, T]$ for all large n . Let

$$\zeta_n := \inf\{t \geq t_1 : q_n(t, \ell) = 1\}.$$

We claim that $\zeta_n \leq t_1 + R$ for all large n . If not, then $q_n(t, \ell) < 1$ on $[t_1, t_1 + R)$, while (7) and no-saturation hold. Applying Lemma 4.4 with $J = \ell$, $s = t_1$ and $t = t_1 + R$ yields

$$C_{n,\ell}(t_1 + R) \geq C_{n,\ell}(t_1) + a_*R - 2e_n > \ell$$

for all large n , because $C_{n,\ell}(t_1) \geq 0$ and $a_*R = \ell + 1$. This is impossible, since $C_{n,\ell} \leq \ell$. Hence $\zeta_n \leq t_1 + R$.

At time ζ_n , (7) gives $\ell_n(\zeta_n) \leq \ell$. If $\ell_n(\zeta_n) < \ell$, then $h_n(\zeta_n) = \ell_n(\zeta_n) + \Delta \leq \ell$, and $q_n(\zeta_n, h_n(\zeta_n)) = 1$, contradicting no-saturation. Thus

$$\ell_n(\zeta_n) = \ell. \quad (8)$$

It remains to prove that the threshold cannot subsequently leave ℓ . By (6), increases from ℓ are impossible on $[t_1, T]$, because $q_n(t, \ell + \Delta) < 1 - 1/n$. The only possible way to leave ℓ is therefore a decrease, which can occur only if $q_n(t, \ell) \leq \alpha_n$ just before an arrival. Suppose that this happens, and let

$$\tau_n := \inf\{t \geq \zeta_n : q_n(t, \ell) \leq \alpha_n\}.$$

Let s_n be the last exit time from the full ℓ th column before τ_n : more precisely, s_n is the last time in $[\zeta_n, \tau_n]$ at which a service exit changes $q_n(\cdot, \ell)$ from 1 to a value strictly below 1. Such a time exists on the standard construction because event times are isolated and $q_n(\zeta_n, \ell) = 1$ whereas $q_n(\tau_n, \ell) \leq \alpha_n < 1$. After this time and before τ_n the ℓ th column is never full. Moreover,

$$C_{n,\ell}(s_n) \geq \ell - \frac{1}{n},$$

since, at the last exit from the full column, one service completion can reduce $C_{n,\ell}$ by at most $1/n$. The threshold is still ℓ on $[s_n, \tau_n]$, and no-saturation holds there. Applying Lemma 4.4 on $[s_n, \tau_n]$ gives

$$C_{n,\ell}(\tau_n) \geq \ell - \frac{1}{n} - 2e_n.$$

But $q_n(\tau_n, \ell) \leq \alpha_n$ implies

$$C_{n,\ell}(\tau_n) = \sum_{i=1}^{\ell} q_n(\tau_n, i) \leq (\ell - 1) + \alpha_n = \ell - (1 - \alpha_n).$$

This contradicts $e_n = o(1 - \alpha_n)$ and $1/n = o(1 - \alpha_n)$ for all large n . Therefore no such τ_n exists, and $\ell_n(t) = \ell$ for all $t \in [\zeta_n, T]$.

Since $\zeta_n \leq t_1 + R$, the theorem follows with any deterministic

$$\sigma > t_1 + R.$$

The constants t_0, t_1, R, σ do not depend on the final horizon T ; only the almost-sure index after which the conclusion holds may depend on T . $\square$

Remark 4.5 (Why the extra condition appears). *The proof needs a positive drift of the first ℓ occupancy columns whenever the ℓ th column is not full. Arrivals add mass to those columns at rate λ , while service exits from pools with at most ℓ tasks can remove mass at rate at most $\beta\ell$. The condition $\beta\ell < \lambda$ makes this comparison pathwise. For exponential service, $\beta = \mu$ and the condition is just $\mu\ell < \lambda$, equivalently $\ell < \rho$, which is automatic when $\rho \notin \Delta\mathbb{Z}_+$. For a general Coxian law, however, a phase may have a much larger instantaneous exit rate than μ , and the argument above no longer proves positive drift. This is precisely where the present note falls short of the full Coxian/refined extension.*

5 Conclusion

The infinite-horizon counterexample is correct after the fixes above, but it addresses a statement stronger than the authors' finite-horizon theorem. The finite-horizon part of this note proves real partial progress: the missing threshold-equilibrium assertion for the refined scheme with Coxian service times is established under the additional slow-exit condition $\beta[\rho]_\Delta < \lambda$. Combined with the authors' observation that the balance estimates corresponding to (8b) and (8c) can be obtained for the refined scheme once the threshold is controlled, this gives a conditional solution for that subclass. The full Coxian/refined problem, without the slow-exit condition, remains open from the arguments given here.

References

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